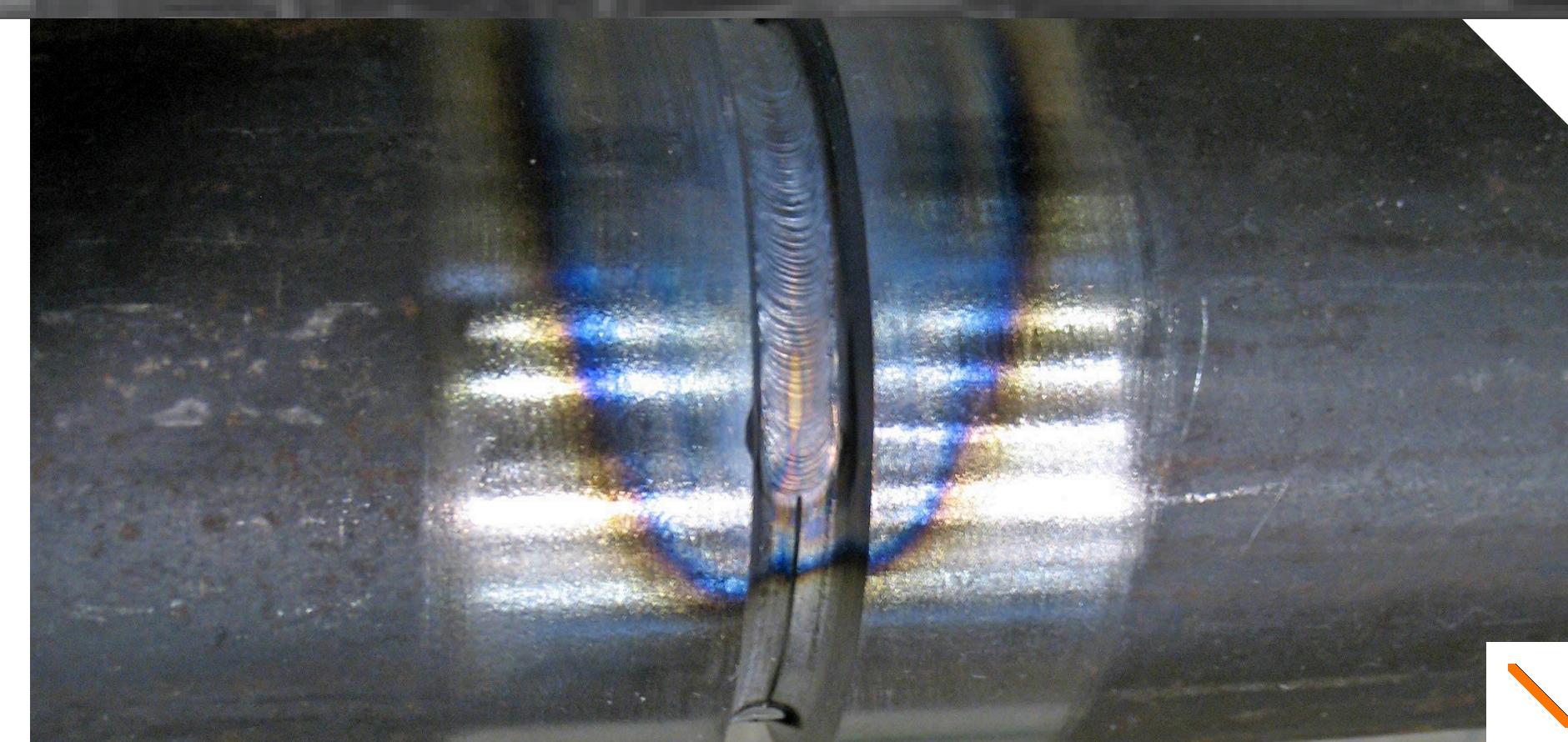


FINAL YEAR ENGINEERING PROJECT

ORBITAL TIG WELDING FOR INDUSTRIAL PIPING

NATHAN JUST

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Polytech Montpellier · Class of 2023–2026
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THE COMPANY

-  **CIMAT SARTEC**, FOSELEV Group. Industrial piping and maintenance.
-  Chemical, petrochemical, manufacturing and steel industries.
-  Boilermaking, steel structures, piping and pressure equipment.
-  1155 boulevard du Grand Terme, 30133 Les Angles.

MY ROLE

Apprentice project engineer: estimating, supplier consultation, planning, fabrication follow-up and inspection, through to installation on site.

ORBITAL TIG WELDING

-  Torch carried right around a pipe held in **fixed position**.
-  **Pulsed current**: the molten pool stays under control.
-  **Four angular sectors**, one set of parameters for each welding position.
-  Enclosed head, argon shielding and **back purging** of the root.

SCOPE STUDIED

Austenitic stainless steel 304L and 316L, small and medium diameters, the everyday production of the workshop.

WHAT WOULD CHANGE

-  **Today**: every pipe weld made by hand, the result tied to the welder.
-  **Tomorrow**: programmed travel, a bead that comes out the same every time.
-  Roughly **half the welding time** on the joint taken as a reference.
-  Real welding parameters **logged automatically**, weld after weld.

COMPLIANCE PATH

Welding procedure qualified first, then the operators to ISO 14732, then the usual inspections. Nothing stands in the way.

WORTH THE INVESTMENT

Welding times to be confirmed by measurement in the workshop before the order is signed.

WHAT COMES NEXT

-  Welding on site, alongside the teams.
-  Open head fed with wire for large diameters and thick walls.
-  A library of welding programmes that gains value over time.

IMAGE CREDITS

Weld bead on 316L stainless steel: 4X4 Blazer 1776, Wikimedia Commons, CC BY 2.0.
Orbital head and welded tube: AliWiki199, Wikimedia Commons, CC BY-SA 4.0.
Pipe weld with heat affected zone: Ytrottier, Wikimedia Commons, CC BY-SA 3.0.
Industrial site: FOSELEV.